# Digital Twin of a Modular Ammonia Cracker for Decentralized Hydrogen Production

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## Abstract

Hydrogen is considered an important energy carrier for future low-carbon energy systems. However, hydrogen storage and transport remain technically challenging because of its low volumetric energy density. Ammonia offers a promising alternative hydrogen carrier because it contains a high hydrogen fraction, can be liquefied more easily than hydrogen, and already has an established global production and transport infrastructure. One possible route for decentralized hydrogen supply is ammonia cracking, where ammonia is decomposed into hydrogen and nitrogen.

This term paper presents the development of a simplified digital twin for a modular ammonia cracker. The model was implemented in Python and visualized using a Streamlit dashboard. The digital twin calculates hydrogen production, nitrogen production, ammonia slip, reaction heat demand, net heat demand after heat recovery, and a basic safety status based on temperature, pressure, and ammonia slip. A process flow diagram and chart analysis were added to improve the technical understanding of the system. The project demonstrates how digital tools can support early-stage process analysis for decentralized hydrogen production.

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## 1. Introduction

The global energy transition requires clean, flexible, and transportable energy carriers. Hydrogen is widely discussed as a key energy carrier because it can be produced from renewable electricity by water electrolysis and used in fuel cells, industrial processes, mobility, and power systems. However, direct hydrogen storage and distribution are difficult because hydrogen has a low volumetric energy density and requires either high-pressure compression, liquefaction at very low temperatures, or chemical storage.

Ammonia is a promising hydrogen carrier because it contains hydrogen in chemically bound form. It can be stored as a liquid under moderate pressure or low temperature, and it has an existing industrial infrastructure due to its use in fertilizer production. Ammonia can be transported to decentralized locations and converted back into hydrogen through ammonia cracking.

The main reaction of ammonia cracking is:

```
2 NH3 → N2 + 3 H2
```

This reaction produces hydrogen and nitrogen without direct carbon dioxide formation. However, ammonia cracking is endothermic, meaning that heat must be supplied to the reactor. In addition, incomplete conversion can lead to ammonia slip, which is important from both safety and purification perspectives.

The aim of this project is to build a simplified digital twin of a modular ammonia cracker for decentralized hydrogen production. The digital twin is not intended for real plant operation, but it provides a conceptual simulation tool for understanding mass balance, energy demand, heat recovery, hydrogen output, and basic safety behavior.

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# 2. Project Objective

The objective of this project is to design and implement a Python-based digital twin for a modular ammonia cracking system. The model is displayed through an interactive Streamlit dashboard.

The main objectives are:

1. To model the stoichiometric conversion of ammonia into hydrogen and nitrogen.
2. To calculate hydrogen production in kg/h and Nm3/h.
3. To calculate nitrogen production and unconverted ammonia slip.
4. To estimate reaction heat demand using a simplified reaction enthalpy.
5. To estimate net heat demand after heat recovery.
6. To include basic safety logic based on temperature, pressure, and ammonia slip.
7. To visualize the process through a process flow diagram and chart analysis.
8. To create a portfolio-style technical project suitable for academic and professional presentation.

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# 3. Background Theory

## 3.1 Ammonia as a Hydrogen Carrier

Ammonia has the chemical formula NH3 and contains three hydrogen atoms per molecule. It is attractive as a hydrogen carrier because it can be liquefied more easily than hydrogen and transported using existing infrastructure. Compared with compressed hydrogen, ammonia can store hydrogen at higher volumetric density under less extreme conditions.

A key advantage of ammonia is that it does not contain carbon. Therefore, when ammonia is cracked into hydrogen and nitrogen, carbon dioxide is not directly produced. This makes green ammonia, produced from renewable hydrogen and nitrogen, a possible energy carrier for future renewable energy systems.

## 3.2 Ammonia Cracking Reaction

The main reaction in an ammonia cracker is:

```
2 NH3 → N2 + 3 H2
```

This means that two moles of ammonia produce one mole of nitrogen and three moles of hydrogen.

From this stoichiometry:

```
1 mol NH3 produces 1.5 mol H2
1 mol NH3 produces 0.5 mol N2
```

This relationship is the foundation of the mass balance used in the digital twin.

### 3.3 Energy Demand

Ammonia cracking is an endothermic reaction. This means heat must be supplied to the reactor for the reaction to proceed. The approximate reaction heat is:

```
ΔH ≈ +92.4 kJ per 2 mol NH3
ΔH ≈ +46.2 kJ per mol NH3
```

In this project, the simplified value of 46.2 kJ/mol NH3 is used to estimate the heat required by the reaction.

The reaction heat demand is calculated from the amount of ammonia converted:

```
Reaction heat = converted NH3 × 46.2 kJ/mol
```

The net heat demand is then calculated by considering heat recovery:

```
Net heat demand = reaction heat × (1 - heat recovery efficiency)
```

### 3.4 Ammonia Slip

Ammonia slip is the amount of ammonia that remains unconverted after the cracking reaction. It is calculated as:

```
NH3 slip = NH3 feed × (1 - conversion)
```

Ammonia slip is important because ammonia is toxic and can also affect downstream hydrogen purification. Therefore, minimizing ammonia slip is important for both process safety and product quality.

### 3.5 Digital Twin Concept

A digital twin is a digital representation of a physical system or process. In industrial applications, digital twins can be used for monitoring, simulation, optimization, predictive maintenance, and operator support.

In this project, the digital twin is a simplified simulation model. It does not include full reactor kinetics, catalyst deactivation, detailed heat transfer, pressure drop, or control-system dynamics. However, it provides a useful conceptual model for understanding the relationship between input parameters and process outputs.

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## 4. Methodology

### 4.1 Software Tools

The project was developed using:

- Python for process calculations
- Streamlit for the interactive web dashboard
- Pandas for data handling
- Plotly or Streamlit charts for visualization
- Draw.io or generated image tools for the process flow diagram
- GitHub and Streamlit Cloud for project sharing and deployment

### 4.2 Model Inputs

The dashboard allows the user to change the following operating parameters:

| Input Parameter | Unit | Description |
|---|---:|---|
| NH3 feed rate | kg/h | Ammonia feed entering the cracker |
| NH3 conversion | % | Fraction of ammonia converted |
| Reactor temperature | °C | Operating temperature of the reactor |
| Pressure | bar | Operating pressure |
| Heat recovery efficiency | % | Fraction of heat demand recovered |
| H2 purification recovery | % | Hydrogen recovered after purification |

### 4.3 Model Outputs

The digital twin calculates:

| Output Parameter | Unit | Description |
|---|---:|---|
| H2 raw production | kg/h | Hydrogen produced before purification |

| Output Parameter | Unit | Description |
|---|---|---|
| H2 product | kg/h | Hydrogen recovered after purification |
| H2 product volume | Nm3/h | Normal volume flow of hydrogen |
| N2 production | kg/h | Nitrogen generated by cracking |
| NH3 slip | kg/h | Unconverted ammonia |
| NH3 slip | % | Ammonia slip relative to feed |
| Reaction heat demand | kW | Heat required for the cracking reaction |
| Net heat demand | kW | Heat demand after heat recovery |
| Safety status | Safe / Warning / Shutdown | Basic process safety classification |

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# 5. Model Development

## 5.1 Mass Balance

The mass balance is based on molecular weights:

```
MW NH3 = 17.031 kg/kmol
MW H2  = 2.016 kg/kmol
MW N2  = 28.014 kg/kmol
```

The ammonia feed in kg/h is converted to kmol/h:

```
NH3 kmol/h = NH3 feed kg/h / MW NH3
```

The converted ammonia is:

```
Converted NH3 = NH3 feed × conversion
```

Using the stoichiometry of the reaction:

```
H2 kmol/h = 1.5 × converted NH3 kmol/h
N2 kmol/h = 0.5 × converted NH3 kmol/h
```

The hydrogen and nitrogen mass flow rates are then calculated using their molecular weights.

### 5.2 Hydrogen Purification

A simplified hydrogen purification recovery is included. The raw hydrogen production is multiplied by the hydrogen recovery efficiency:

```
H2 product = H2 raw × H2 recovery
```

This represents losses in membrane purification, pressure swing adsorption, or another hydrogen separation process.

### 5.3 Heat Demand

The reaction heat is calculated using:

```
46.2 kJ/mol NH3
```

The model converts the heat demand from kJ/h to kW:

```
kW = kJ/h / 3600
```

The net heat demand is calculated using heat recovery efficiency:

```
Net heat demand = reaction heat demand × (1 - heat recovery efficiency)
```

### 5.4 Safety Logic

A basic safety assessment is included in the digital twin. The safety status is based on reactor temperature, pressure, and ammonia slip.

The simplified safety limits are:

| Parameter | Safe Range | Warning Range | Shutdown Range |
|---|---|---|---|
| Temperature | 550–700°C | 500–550°C or 700–750°C | <500°C or >750°C |
| Pressure | $\leq$ 6 bar | 6–8 bar | >8 bar |
| $NH_3$ slip | $\leq$ 1% | 1–5% | >5% |

The dashboard displays one of three operating states:

```
Safe
Warning
Shutdown
```

It also provides the reason and a recommended action.

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## 6. Process Flow Diagram

A process flow diagram was included in the dashboard to improve the understanding of the system. The main process route is:

```
Liquid NH3 Storage
→ NH3 Pump / Feed Valve
→ Evaporator
→ Preheater / Heat Exchanger
→ Ammonia Cracker Reactor
→ Cooler / Heat Recovery
→ NH3 Slip Removal
→ H2 Purification / Membrane
→ H2 Product
```

A side stream from the purification system represents nitrogen-rich off-gas.

The process flow diagram also includes sensor markers:

```
T = Temperature sensor
P = Pressure sensor
NH3 = Ammonia analyzer
H2 = Hydrogen analyzer
```

This makes the dashboard more realistic and connects the process model to industrial monitoring concepts.

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## 7. Dashboard Development

The dashboard was developed using Streamlit. The sidebar contains the user inputs, while the main page displays the results, process diagram, chart analysis, and safety assessment.

The dashboard structure is:

1. Project title and description

2. Main reaction explanation
3. Sidebar input parameters
4. Main simulation results
5. Process flow diagram
6. Chart analysis
7. Safety assessment
8. Model assumptions and limitations

The dashboard provides an interactive environment where users can change process conditions and immediately observe their effect on hydrogen production, ammonia slip, and heat demand.

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## 8. Chart Analysis

Chart analysis was included to improve the engineering interpretation of the digital twin.

### 8.1 Hydrogen Production vs Ammonia Feed Rate

The first chart shows hydrogen production as a function of ammonia feed rate. In the simplified model, this relationship is linear because the stoichiometric mass balance directly links ammonia feed to hydrogen output. When ammonia feed increases, hydrogen production increases proportionally.

This result is expected because:

```
H2 production ∝ NH3 feed rate
```

The chart is useful for estimating hydrogen output for different system sizes.

### 8.2 Ammonia Slip vs Conversion

The second chart shows ammonia slip as a function of ammonia conversion. As conversion increases, ammonia slip decreases. In the simplified model, this relationship is also linear because ammonia slip is calculated directly from unconverted ammonia.

The relationship is:

```
NH3 slip = NH3 feed × (1 - conversion)
```

This chart is useful because ammonia slip is an important safety and purification parameter. Lower ammonia slip improves hydrogen product quality and reduces downstream treatment requirements.

### 8.3 Net Heat Demand vs Heat Recovery Efficiency

The third chart shows net heat demand as a function of heat recovery efficiency. As heat recovery increases, net heat demand decreases. In the simplified model, this relationship is linear because heat recovery is applied as a percentage reduction of reaction heat demand.

The relationship is:

```
Net heat demand = reaction heat × (1 - heat recovery)
```

This chart highlights the importance of heat recovery in ammonia cracking systems. Since the reaction is endothermic, efficient heat recovery can reduce the external energy required by the process.

### 8.4 Temperature-Based Conversion

A more realistic improvement is the use of temperature-based conversion. In a real ammonia cracker, conversion depends strongly on reactor temperature, catalyst properties, pressure, and residence time. A simplified temperature-conversion relationship can be introduced:

| Reactor Temperature | Estimated NH3 Conversion |
|---:|---:|
| 500°C | 70% |
| 550°C | 85% |
| 600°C | 95% |
| 650°C | 98% |
| 700°C | 99% |
| 750°C | 99.5% |

This creates a non-linear trend where conversion increases strongly with temperature and then approaches a maximum value. This improvement makes the dashboard more realistic than a purely linear model.

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## 9. Results and Discussion

The digital twin successfully calculates the main performance indicators of a modular ammonia cracker. The results show that hydrogen production increases with ammonia feed rate and conversion. The model also shows that incomplete conversion creates ammonia slip, which can lead to safety and purification concerns.

The heat demand calculation confirms that ammonia cracking requires external energy because the reaction is endothermic. Heat recovery significantly reduces the net heat demand, making it an important design consideration for modular ammonia crackers.

The safety logic allows the dashboard to classify operating conditions as safe, warning, or shutdown. This improves the educational value of the project because it connects numerical simulation with operational decision-making.

The process flow diagram improves the clarity of the dashboard by showing how ammonia moves through storage, pumping, evaporation, preheating, cracking, cooling, ammonia slip removal, and hydrogen purification. This makes the digital twin more understandable for users who are not familiar with the process.

Some charts are linear because the current model uses simplified stoichiometric and percentage-based equations. This is acceptable for an early-stage conceptual model. However, the addition of temperature-dependent conversion introduces more realistic non-linear behavior.

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## 10. Model Assumptions

The model is based on the following assumptions:

1. The main reaction is only ammonia cracking:

```
2 NH3 → N2 + 3 H2
```

1. The mass balance is based on ideal stoichiometry.

2. Reaction heat is estimated using:

```
46.2 kJ/mol NH3
```

1. Heat recovery is treated as a simple percentage reduction.

2. Hydrogen purification recovery is treated as a fixed percentage.

3. Ammonia slip is calculated only from unconverted ammonia.

4. Reactor temperature affects safety status and can also be used to estimate conversion.

5. The model does not include detailed catalyst kinetics, residence time, pressure drop, detailed thermodynamics, or dynamic control behavior.

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## 11. Limitations

Although the dashboard provides useful insight, it has several limitations:

1. It is not a real plant control system.
2. It is not suitable for safety certification.
3. It does not include detailed reactor kinetics.
4. It does not model catalyst deactivation.
5. It does not include pressure drop calculations.
6. It does not include detailed heat-transfer design.
7. It does not include dynamic start-up or shutdown behavior.
8. It does not include real sensor data or industrial communication protocols.

The model should therefore be understood as a conceptual digital twin for learning, demonstration, and early-stage process analysis.

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## 12. Future Improvements

Several future improvements could make the digital twin more advanced:

1. Add detailed reactor kinetics for ammonia decomposition.
2. Include catalyst type and catalyst deactivation.
3. Model the effect of residence time and reactor volume.
4. Include pressure effects on conversion.
5. Add real-time sensor data input.
6. Add dynamic simulation for start-up and shutdown.
7. Include cost analysis and hydrogen production cost.
8. Add carbon footprint comparison with other hydrogen carriers.
9. Add a control system for temperature and feed regulation.
10. Connect the dashboard to a database for historical data storage.

These improvements would move the project closer to a real industrial digital twin.

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## 13. Conclusion

This project developed a simplified digital twin of a modular ammonia cracker for decentralized hydrogen production. The model calculates hydrogen output, nitrogen production, ammonia slip, reaction heat demand, net heat demand, and safety status. The dashboard was implemented in Python using Streamlit and includes an interactive user interface, process flow diagram, chart analysis, and safety assessment.

The project demonstrates the potential of ammonia as a hydrogen carrier and shows how digital tools can support early-stage process understanding. Although the model is simplified, it provides a strong foundation for future development. With additional kinetic modeling, real sensor data, and dynamic simulation, the digital twin could be extended toward more realistic industrial applications.

Overall, the project combines chemical engineering principles, hydrogen technology, process simulation, and digitalization. It is suitable as an academic term project and as a portfolio project for applications in hydrogen, Power-to-X, renewable fuels, and process engineering.

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